

Observation Task 1

1. Student Utility Program

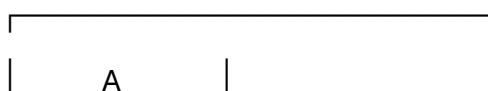
Steps

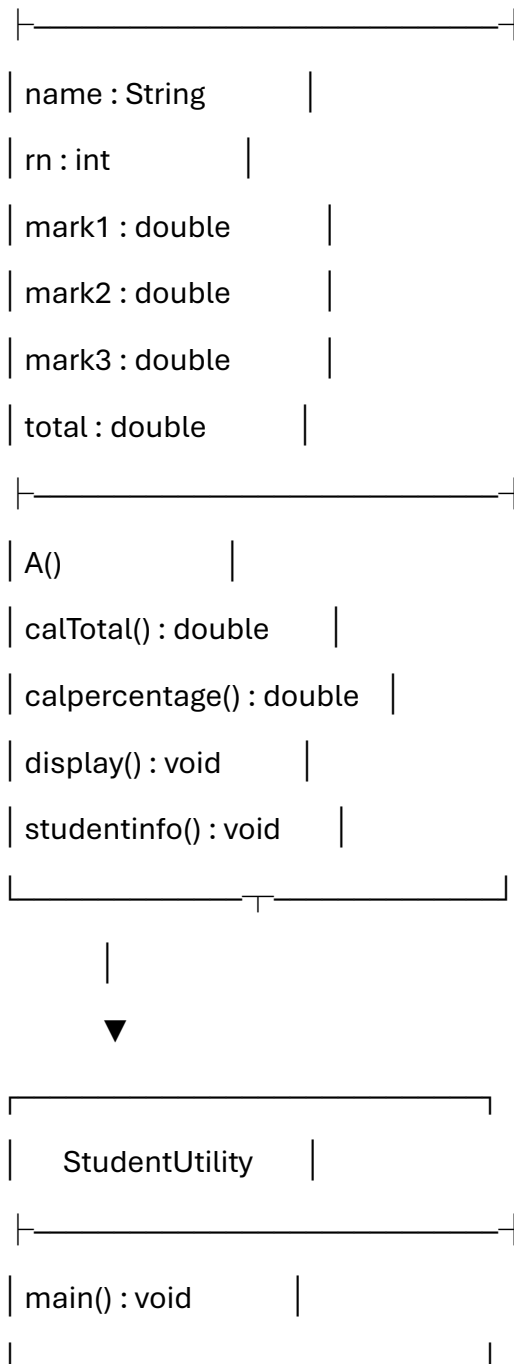
1. Create a class A with student details.
2. Declare name, roll number, three marks, and total.
3. Create methods to calculate total and percentage.
4. Create methods to display student information and result.
5. In the main class, take student details as input.
6. Display a menu with different operations.
7. Perform the selected operation.
8. Repeat the menu until the user chooses Exit.

Algorithm

1. Start.
2. Create class A.
3. Create an object of class A.
4. Read student name, roll number, and three marks.
5. Display the menu.
6. Read the user's choice.
7. If choice is 1, display student information.
8. If choice is 2, calculate and display total and percentage.
9. If choice is 3, display the complete result.
10. If choice is 4, exit the program.
11. Otherwise, display invalid choice.
12. Stop.

UML Diagram





Code

```
import java.util.Scanner;
```

```
class A
```

```
{
```

```
    String name;
```

```
    int rn;
```

```
double mark1, mark2, mark3;
```

```
double total;
```

```
public A()
```

```
{
```

```
    this.name = "";
```

```
    this.rn = 0;
```

```
    this.mark1 = 0;
```

```
    this.mark2 = 0;
```

```
    this.mark3 = 0;
```

```
}
```

```
double calTotal()
```

```
{
```

```
    total = mark1 + mark2 + mark3;
```

```
    return total;
```

```
}
```

```
double calpercentage()
```

```
{
```

```
    double percentage = (total / 300) * 100;
```

```
    return percentage;
```

```
}
```

```
void display()
```

```
{
```

```
    System.out.println("student name: " + name);
```

```
    System.out.println("student roll number: " + rn);
```

```
System.out.println("marks in 1st subject: " + mark1);  
System.out.println("marks in 2nd subject: " + mark2);  
System.out.println("marks in 3rd subject: " + mark3);  
System.out.println("total in all subjects: " + calTotal());  
System.out.println("percentage: " + calpercentage() + "%");  
}
```

```
void studentinfo()  
{  
    System.out.println("student name: " + name);  
    System.out.println("student roll number: " + rn);  
    System.out.println("marks in 1st subject: " + mark1);  
    System.out.println("marks in 2nd subject: " + mark2);  
    System.out.println("marks in 3rd subject: " + mark3);  
}  
}
```

```
public class StudentUtility  
{  
    public static void main(String[] args)  
    {  
        Scanner sc = new Scanner(System.in);  
  
        A obj = new A();  
        System.out.println("Enter name of student:");  
        obj.name = sc.nextLine();  
  
        System.out.println("Enter roll number:");
```

```
obj.rn = sc.nextInt();
```

```
System.out.println("Enter marks of three subjects:");
```

```
obj.mark1 = sc.nextDouble();
```

```
obj.mark2 = sc.nextDouble();
```

```
obj.mark3 = sc.nextDouble();
```

```
int choice = 0;
```

```
while(choice != 4)
```

```
{
```

```
    System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
```

```
    System.out.println("1. Print student details");
```

```
    System.out.println("2. Calculate total and percentage");
```

```
    System.out.println("3. Print result");
```

```
    System.out.println("4. Exit");
```

```
    System.out.print("Enter your choice: ");
```

```
    choice = sc.nextInt();
```

```
    switch(choice)
```

```
    {
```

```
        case 1:
```

```
            obj.studentinfo();
```

```
            break;
```

```
        case 2:
```

```
            System.out.println("Total = " + obj.calTotal());
```

```
        System.out.println("Percentage = " + obj.calpercentage() + "%");

        break;

    case 3:

        obj.display();

        break;

    case 4:

        System.out.println("Exiting program...");

        break;

    default:

        System.out.println("Invalid choice");

    }

}

    sc.close();

}
```

Sample Output

Enter name of student:

Naveen

Enter roll number:

101

Enter marks of three subjects:

85

90

78

===== STUDENT UTILITY PROGRAM =====

1. Print student details
2. Calculate total and percentage
3. Print result
4. Exit

Enter your choice: 3

student name: Naveen

student roll number: 101

marks in 1st subject: 85.0

marks in 2nd subject: 90.0

marks in 3rd subject: 78.0

total in all subjects: 253.0

percentage: 84.33333333333333%

2. Conditional Statement Program

Steps

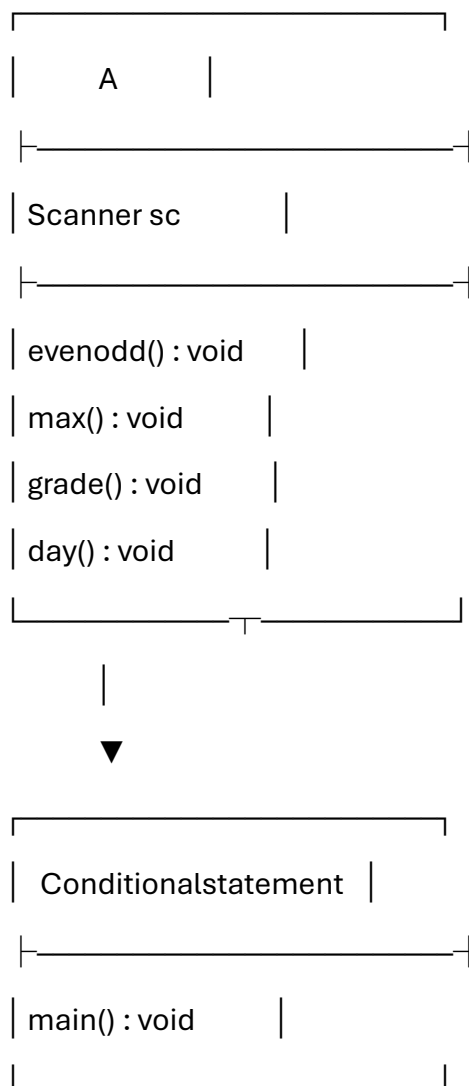
1. Create a class A.
2. Create methods for even/odd, maximum, grade, and day.
3. Take the user's choice.
4. Call the corresponding method.
5. Display the result.

Algorithm

1. Start.
2. Create class A.
3. Create an object of A.
4. Display the menu.

5. Read the choice.
6. If choice is 1, check even or odd.
7. If choice is 2, find the largest of three numbers.
8. If choice is 3, find the grade.
9. If choice is 4, display the day.
10. Otherwise, display invalid choice.
11. Stop.

UML Diagram



Code

```
import java.util.Scanner;
```

```
class A {  
    Scanner sc = new Scanner(System.in);  
  
    void evenodd() {  
        System.out.print("Enter number: ");  
        int n = sc.nextInt();  
  
        if (n % 2 == 0) {  
            System.out.println("n is even");  
        } else {  
            System.out.println("n is odd number");  
        }  
    }  
}  
  
    void max() {  
        System.out.print("Enter three numbers: ");  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        int c = sc.nextInt();  
  
        if (a >= b && a >= c) {  
            System.out.println("a is max: " + a);  
        } else if (b >= a && b >= c) {  
            System.out.println("b is max: " + b);  
        } else {  
            System.out.println("c is max: " + c);  
        }  
    }  
}
```

```
void grade() {  
    System.out.print("Enter percentage: ");  
    int per = sc.nextInt();  
  
    if (per >= 90 && per <= 100) {  
        System.out.println("A grade");  
    } else if (per >= 75) {  
        System.out.println("B grade");  
    } else if (per >= 50) {  
        System.out.println("C grade");  
    } else if (per >= 35) {  
        System.out.println("D grade");  
    } else {  
        System.out.println("Fail");  
    }  
}
```

```
void day() {  
    System.out.println("Enter number from 1-7: ");  
    int choice = sc.nextInt();  
  
    switch (choice) {  
        case 1:  
            System.out.println("Sunday");  
            break;  
        case 2:  
            System.out.println("Monday");  
    }
```

```
        break;
    case 3:
        System.out.println("Tuesday");
        break;
    case 4:
        System.out.println("Wednesday");
        break;
    case 5:
        System.out.println("Thursday");
        break;
    case 6:
        System.out.println("Friday");
        break;
    case 7:
        System.out.println("Saturday");
        break;
    default:
        System.out.println("Invalid choice");
    }
}
}
```

```
public class Conditionalstatement {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
A obj = new A();
```

```
System.out.println("1. Even or Odd");
```

```
System.out.println("2. Largest of Three Numbers");
```

```
System.out.println("3. Grade");
```

```
System.out.println("4. Day of the Week");
```

```
System.out.print("Enter your choice: ");
```

```
int choice = sc.nextInt();
```

```
switch (choice) {
```

```
    case 1:
```

```
        obj.evenodd();
```

```
        break;
```

```
    case 2:
```

```
        obj.max();
```

```
        break;
```

```
    case 3:
```

```
        obj.grade();
```

```
        break;
```

```
    case 4:
```

```
        obj.day();
```

```
        break;
```

```
    default:
```

```
        System.out.println("Invalid choice");
```

```
}
```

```
sc.close();
```

```
}  
  
}
```

Sample Output

1. Even or Odd
2. Largest of Three Numbers
3. Grade
4. Day of the Week

Enter your choice: 2

Enter three numbers: 12 25 7

b is max: 25

3. Looping Statement Program

Steps

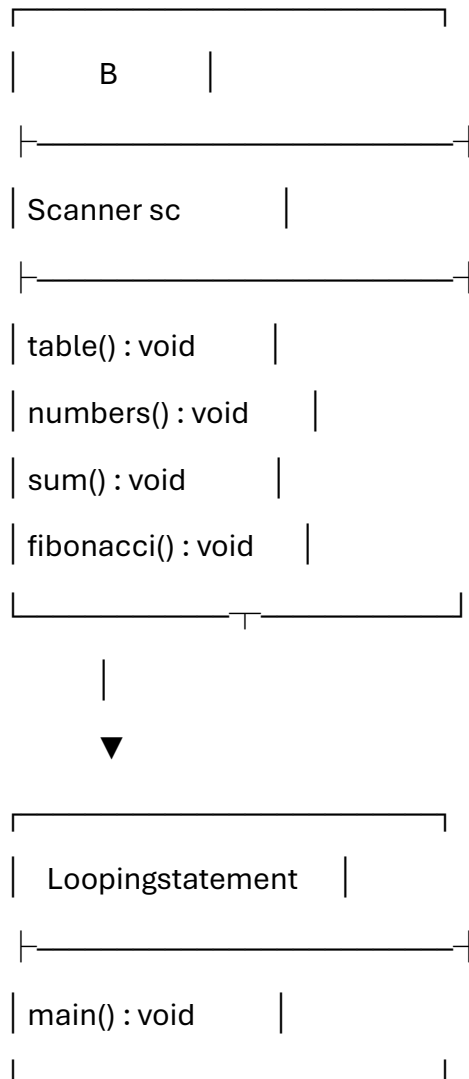
1. Create a class B.
2. Create methods for multiplication table, numbers, sum, and Fibonacci series.
3. Take the user's choice.
4. Call the selected method.
5. Display the result.

Algorithm

1. Start.
2. Create class B.
3. Create an object of B.
4. Display the menu.
5. Read the choice.
6. If choice is 1, print multiplication table.
7. If choice is 2, print numbers from 1 to N.
8. If choice is 3, calculate sum of first N natural numbers.

9. If choice is 4, print Fibonacci series.
10. Otherwise, display invalid choice.
11. Stop.

UML Diagram



Code

```
import java.util.Scanner;
```

```
class B {
```

```
    Scanner sc = new Scanner(System.in);
```

```
void table() {  
    System.out.print("Enter a number: ");  
    int n = sc.nextInt();  
  
    for (int i = 1; i <= 10; i++) {  
        System.out.println(n + " x " + i + " = " + (n * i));  
    }  
}
```

```
void numbers() {  
    System.out.print("Enter N: ");  
    int n = sc.nextInt();  
  
    for (int i = 1; i <= n; i++) {  
        System.out.println(i);  
    }  
}
```

```
void sum() {  
    System.out.print("Enter N: ");  
    int n = sc.nextInt();  
  
    int sum = 0;  
  
    for (int i = 1; i <= n; i++) {  
        sum = sum + i;  
    }  
}
```

```

        System.out.println("Sum = " + sum);
    }

    void fibonacci() {
        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();

        int a = 0;
        int b = 1;

        System.out.print("Fibonacci Series: ");

        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");

            int c = a + b;
            a = b;
            b = c;
        }

        System.out.println();
    }
}

public class Loopingstatement {

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);
```

```
B obj = new B();
```

```
System.out.println("1. Multiplication Table");
```

```
System.out.println("2. Display Numbers from 1 to N");
```

```
System.out.println("3. Sum of First N Natural Numbers");
```

```
System.out.println("4. Fibonacci Series");
```

```
System.out.print("Enter your choice: ");
```

```
int choice = sc.nextInt();
```

```
switch (choice) {
```

```
    case 1:
```

```
        obj.table();
```

```
        break;
```

```
    case 2:
```

```
        obj.numbers();
```

```
        break;
```

```
    case 3:
```

```
        obj.sum();
```

```
        break;
```

```
    case 4:
```

```
        obj.fibonacci();
```

```
        break;

    default:

        System.out.println("Invalid choice");
    }

    sc.close();
}
}
```

Sample Output

1. Multiplication Table
2. Display Numbers from 1 to N
3. Sum of First N Natural Numbers
4. Fibonacci Series

Enter your choice: 4

Enter number of terms: 7

Fibonacci Series: 0 1 1 2 3 5 8

4. Methods Program

Steps

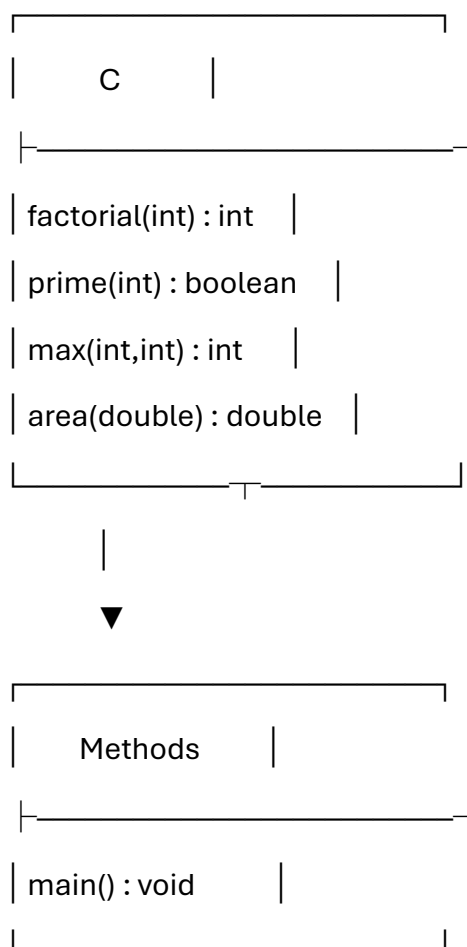
1. Create a class C.
2. Create a method to calculate factorial.
3. Create a method to check whether a number is prime.
4. Create a method to find the maximum of two numbers.
5. Create a method to calculate the area of a circle.
6. Create an object of class C.
7. Take input from the user.

8. Call each method and display the result.

Algorithm

1. Start.
2. Create class C.
3. Create an object of C.
4. Read a number and call the factorial method.
5. Read a number and call the prime method.
6. Read two numbers and call the maximum method.
7. Read radius and call the area method.
8. Display all results.
9. Stop.

UML Diagram



Code

```
import java.util.Scanner;
```

```
class C {
```

```
    int factorial(int n) {
```

```
        int fact = 1;
```

```
        for (int i = 1; i <= n; i++) {
```

```
            fact = fact * i;
```

```
        }
```

```
        return fact;
```

```
    }
```

```
    boolean prime(int n) {
```

```
        if (n <= 1) {
```

```
            return false;
```

```
        }
```

```
        for (int i = 2; i < n; i++) {
```

```
            if (n % i == 0) {
```

```
                return false;
```

```
            }
```

```
        }
```

```
        return true;
```

```
    }
```

```
int max(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```

```
double area(double r) {  
    return Math.PI * r * r;  
}  
}
```

```
public class Methods {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        C obj = new C();
```

```
        System.out.print("Enter number for factorial: ");
```

```
        int n = sc.nextInt();
```

```
        System.out.println("Factorial = " + obj.factorial(n));
```

```
        System.out.print("Enter number to check prime: ");
```

```
        int p = sc.nextInt();
```

```
if (obj.prime(p)) {  
    System.out.println(p + " is prime");  
} else {  
    System.out.println(p + " is not prime");  
}  
  
System.out.print("Enter two numbers: ");  
int a = sc.nextInt();  
int b = sc.nextInt();  
System.out.println("Maximum = " + obj.max(a, b));  
  
System.out.print("Enter radius: ");  
double r = sc.nextDouble();  
System.out.println("Area of circle = " + obj.area(r));  
  
sc.close();  
}  
}
```

```
PS C:\Users\navee\OneDrive\Desktop\programs> java StudentUtility
```

```
Enter name of student:
```

```
naveen
```

```
Enter roll number:
```

```
248
```

```
Enter marks of three subjects:
```

```
30
```

```
70
```

```
60
```

```
===== STUDENT UTILITY PROGRAM =====
```

```
1. Print student details
```

```
2. Calculate total and percentage
```

```
3. Print result
```

```
4. Exit
```

```
Enter your choice: 3
```

```
student name: naveen
```

```
student roll number: 248
```

```
marks in 1st subject: 30.0
```

```
marks in 2nd subject: 70.0
```

```
marks in 3rd subject: 60.0
```

```
total in all subjects: 160.0
```

```
percentage: 53.333333333333336%
```

```
PS C:\Users\navee\OneDrive\Desktop\programs> java Condition
```

```
1. Even or Odd
```

```
2. Largest of Three Numbers
```

```
3. Grade
```

```
4. Day of the Week
```

```
Enter your choice: 1
```

```
Enter number: 99
```

```
n is odd number
```

```
PS C:\Users\navee\OneDrive\Desktop\programs> java Loopingstatement
1. Multiplication Table
2. Display Numbers from 1 to N
3. Sum of First N Natural Numbers
4. Fibonacci Series
Enter your choice: 1
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

```
PS C:\Users\navee\OneDrive\Desktop\programs> javac Methods.java
PS C:\Users\navee\OneDrive\Desktop\programs> java Methods
Enter number for factorial: 5
Factorial = 120
Enter number to check prime: 6
6 is not prime
Enter two numbers: 5
8
Maximum = 8
Enter radius: 6
Area of circle = 113.09733552923255
```